

DOLITRAIL · PRECIS

DoliTrail design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	TRL 2 and 3 update; constructable kit (DLT-DDR-001, DLT-DDR-002), components with BOM numbers, first-order numbers, safety
0.1	2026-09-30	Amish Chadha	Initial scaffold

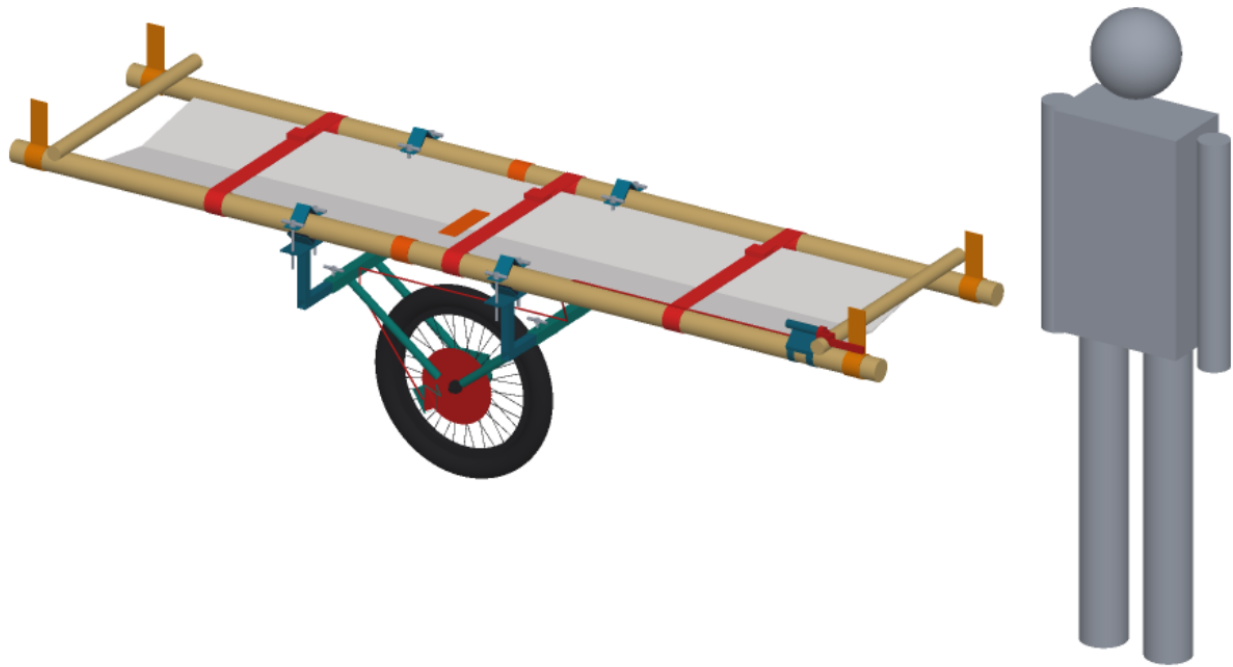
DoliTrail design precis

Adds a braked wheel and harness to the bamboo stretchers families use to carry patients to the road.

DoliTrail

github.com/BoujeeEnjinia1701/dolitrail

CONCEPT, NOT FOR FABRICATION



Kit fitted to the reference doli; seen from the front right and above, 24 deg elevation. Grey figure: 1.75 m person for scale

Figure 1. The kit fitted to the reference doli, with a 1.75 m person for scale (concept render).

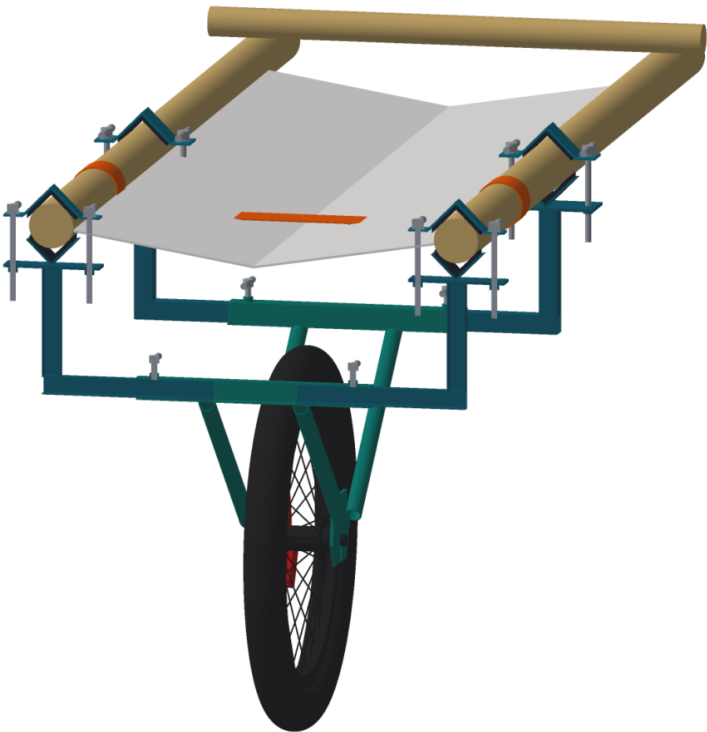
■ How it works

DoliTrail is a clamp-on kit for the doli a family has already built. A single 20 inch wheel runs under the patient's hips in a welded steel fork unit. Four side arms slide into the fork unit's two cross stubs, set to the doli's pole spacing, and each ends in a rubber-lined steel V that the bamboo pole sits in; an upper jaw and two T-handle bolts close each clamp by hand. A cross strap round both poles above the wheel keeps the bed from sagging onto the tyre. A disc brake on the wheel is worked by a lever with a parking lock, strapped to the front right pole beside the front bearer's hand. Each bearer wears a harness whose slings take the pole ends at knuckle height, so on level ground and gentle slopes the wheel carries most of the weight and the bearers balance and steer. On descents the front bearer holds speed with the brake. At steps, rocks and streams the bearers lift the doli and carry it, as they do today.

DoliTrail: cutaway

github.com/BoujeeEnjinia1701/dolitrail

CONCEPT, NOT FOR FABRICATION



Cut across the doli through the front pair of clamps; seen from the front and slightly to the left, 10 deg elevation. Each pole sits in a rubber-lined steel V, closed by a jaw and two T-bolts; the wheel runs between the cross stubs, below the cross strap and the bed

Figure 2. Cut across the doli through the front pair of clamps.

■ Components

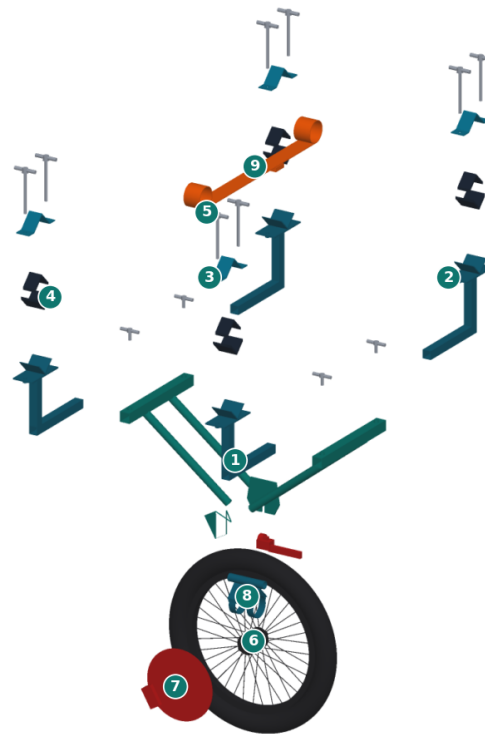
Table 1. Components (numbers are BOM lines in bom/bom.csv).

#	Component	Role
1	Fork unit	Welded steel: two 30 mm square cross stubs, four 22 mm struts, dropout plates with the brake caliper lobe
2	Side arms (4)	Square tube spigot that slides in a stub, a post and a 90 deg V saddle under the pole
3	Upper jaws (4)	Bent 3 mm steel V over the pole, with two ears for the bolts
4	Jaw liners (8)	5 mm rubber in each saddle and jaw: grip on wet bamboo and no crushing points
5	T-handle bolts (8) and set screws (4)	Close the clamps and lock the arms by hand; no loose tools
6	Wheel	20 inch cargo or tricycle wheel, 57-406 knobbly tyre, disc hub, solid axle
7	Disc brake set	203 mm rotor, cable caliper with sintered pads, lever with parking lock, cable
8	Lever mount	Rubber-lined V with a 22.2 mm bar stub, strapped to the front right pole
9	Cross strap	50 mm webbing loop round both poles above the wheel, under the bed
10	Patient straps (3)	Chest, hips and legs; cam buckles with a pull tab
11	Bearer harnesses (2)	Hip belt, padded shoulder yoke and two adjustable slings ending in pole loops
12 to 15	Bag, fitting card, cable ties, pump and puncture kit	Carry, fit and keep the kit running

DoliTrail: exploded viewgithub.com/BoujeeEnjinia1701/dolitrail

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- 1 Fork unit
- 2 Side arms (4)
- 3 Upper jaws (4)
- 4 Jaw liners (8)
- 5 T-handle bolts (8) and set screws (4)
- 6 Wheel
- 7 Disc brake set
- 8 Lever mount
- 9 Cross strap



Frame, wheel and brake seen from the front right and above, 22 deg elevation; numbers match bom/bom.csv. Lever mount drawn near the frame (it fits on the front right pole). Not shown: brake cable, patient straps (10), bearer harnesses (11)

Figure 3. Exploded view of the frame, wheel and brake; numbers match the BOM.

■ Key design choices

- **One wheel under the hips** (DLT-DDR-001, D1). The wheel stays inside the doli's own width; a pair would need a track wider than most trails.
- **Clamp on, adjust by hand** (D2, DLT-DDR-002). Telescoping arms reach poles 450 to 600 mm apart; V saddles self-centre poles of 40 to 80 mm; T-handles mean no spanner is needed.
- **Disc brake with a parking lock** (D3). Works when wet better than a rim brake on a muddy rim; the lock holds the doli when the bearers rest on a slope.
- **Slings, not cups** (D4). A webbing pole loop fits any pole end; the scaffold's "pole-end cups" became loops.
- **Ordinary parts**. Bicycle and cycle-rickshaw parts, mild steel tube a village workshop can weld, webbing a tailor can sew.

■ First-order numbers

All from DLT-CAL-001; estimates are marked.

Table 2. Key figures.

Quantity	Value
Rolling weight: 120 kg patient, 10 kg doli, 12.3 kg fitted kit	142 kg
Load per bearer on a level trail (estimate)	about 14 kg, against 65 kg carried by two bearers (79 per cent less)
Front bearer on a 30 per cent descent, brake on (estimate)	about 46 kg
Brake torque factor on a 30 per cent grade, wet	1.23
Tyre grip factor on a 30 per cent grade: wet ramp; wet clay	1.44; 0.84
Clamp slip resistance, one pole, wet	3.96 kN (factor 2.6 on 1.5 kN)
Lowest structural factor on yield at twice the rated load	2.02 (struts)
Width on the reference doli	680 mm
Kit mass in the bag; fitted	14.8 kg; 12.3 kg
Fitting time, two people (estimate)	about 4.7 min
Estimated parts cost	USD 455 (value-engineering target USD 1,500)

DoliTrail: where the weight goes on a level trail (all values are estimates)

github.com/BoujeeEnjinia1701/dolitrail

CONCEPT, NOT FOR FABRICATION



Figure 4. Where the weight goes on a level trail (estimates).

■ Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice), kept unchanged:

- Use DoliTrail's own clamp geometry and colours, distinct from the CMC or Ferno Mule II litter wheel. The DoliTrail clamp is a rubber-lined 90 deg V with a separate jaw and two hand bolts, on telescoping square arms; the colours are teal and blue.
- Confirm that US7922183B2 (Hill-Rom stretcher brake) has lapsed for non-payment of fees before using a similar brake arrangement. DoliTrail uses an ordinary bicycle disc brake on the wheel hub, not a stretcher-mounted brake; the check is still listed in DLT-DEC-001.

■ Shared blocks

- CalRig proof-load for clamp, fork, wheel and harness load tests (TRL 4).

■ Safety

Safety: Safety-critical patient transport equipment. Published as an open engineering reference, never as certified medical or rescue equipment.

Brake failure or tyre slip on a steep wet descent could lead to a runaway or a fall. Bearers keep hold of the poles at all times, and lift and carry on wet slopes steeper than the tested limit.

The doli rolls on one wheel and can tip sideways. Both bearers steady it with both hands; on a 15 per cent side slope each hand holds about 16 kg up or down.

Lift and carry across streams and rivers; never roll the doli through moving water.

Check every clamp for grip on every fitting, since bamboo varies in size and can split. Tighten by hand only.

Patient straps must release with one pull so the patient can be moved off the doli at once.

No patient is carried until the frame, clamps and wheel have been proof-loaded (DLT-BLD-001, sections 5 and 6).

This design is published as an open engineering reference. It is not certified equipment.